

Proposed Requirements for a Blockchain-Based Electronic Laboratory Notebook

Alexander Yu Tang¹, Maurice Han Tong Ling^{1,2*}

¹HOHY PTE LTD, Singapore.

²Newcastle Australia Institute of Higher Education, University of Newcastle, Australia.

Abstract

Electronic Laboratory Notebooks (ELNs) is used to support experimental documentation, data management in a collaborative landscape while adhering to regulatory requirements in a centrally controlled system. The use of blockchain technology could provide cryptographically verifiable data, evidence and subsequent alterations. However, the use of blockchain in ELN has not been clearly defined. This study aimed to identify requirements for a blockchain-based ELN. A preliminary scoping review on ELN requirements via PubMed search was conducted for publications indexed on or before 30 June 2026. Forty-one records were identified; of which, 11 were included. The identified requirements were grouped into three categories: system and compliance, data capture and data usage. These findings were integrated with blockchain-specific considerations. A hybrid architecture is proposed comprising an ELN application layer, an encrypted off-chain repository, a blockchain integrity layer, an identity and key-management service and an interoperability layer. Twenty-eight requirement domains are presented, including source data capture, cryptographic hashing, digital signatures, trusted timestamps, role-based access, disaster recovery and governance. Blockchain should supplement rather than replace conventional ELN infrastructure.

Keywords

Electronic Laboratory Notebook (ELN), Blockchain, Research Data Management, Data Integrity, Scientific Recordkeeping, Digital Signatures, Regulatory Compliance, Laboratory Informatics.

Introduction

Whenever erroneous data, including fabricated or falsified data, makes into a scientific publication; it commonly leads to retraction. Fang and Casadevall [1] reports a COPE survey of Medline retractions from 1988 to 2004 and finds 28% of the retractions are due to data fabrication or falsification. A later study by the same group [2] reviewing 2047 retractions on 3 May 2012 finds 67.4% of the retractions are due to misconduct, and this includes 43.4% of the retractions are due to fraud or suspected fraud. This is supported studies showing 66.4% [3] to 72.5% [4] of the retractions are due to data issues, which includes falsification, fabrication, and unreliable results. At the same time, meta-analyses show that between 14.15% [5] and 17.5% [6] of scientists having knowledge of colleague fabricating or falsifying data between 1987 and 2018, which is consistent with the findings in a meta-analysis reported in 2023 on research misconduct [7]. Despite varying prevalence, these studies nevertheless suggest a substantial problem with scientific honesty.

To address this issue, publication of laboratory notebooks especially electronic laboratory notebook (ELN), has been proposed to promote scientific honesty [8-10], on top of third-party notarization [11]. Research / laboratory notebook is a crucial artefact of scientific work and the importance of good recordkeeping plays a pivotal role [12]. Therefore, this study aimed to identify requirements and implementation considerations reported for conventional ELN and to utilise findings with blockchain-specific integrity, privacy, security and governance considerations, to propose an evidence-informed framework for a blockchain-based ELN.

Scoping Review on the Requirements and Considerations for ELN

To enumerate the current requirements and/or considerations for implementing an ELN, a scoping review was conducted using PRISMA methodology [13,14]. PubMed was searched for publications on or before 30 June 2026 using the following search term (URL #1): (require* OR consider*) AND ("electronic laboratory notebook" OR "electronic laboratory notebooks" OR "electronic lab notebook" OR "electronic lab notebooks"). A total of 41 articles were identified. 30 articles were excluded due to (a) non-primary articles (n = 6), and (b) no clear or enumeration of ELN requirements (n = 24). The requirements and/or considerations from the included 11 articles are enumerated in Table 1 below.

Collectively, the main requirements of ELN can be categorized into 3 categories; namely, system and compliance, data capture, and data

Correspondence to: Maurice Han Tong Ling, Newcastle Australia Institute of Higher Education, University of Newcastle, Australia.

Received: 17 Jun 2026; **Accepted:** 02 Aug 2026; **Published:** 10 Aug 2026

Citation: Alexander Yu Tang, Maurice Han Tong Ling. Proposed Requirements for a Blockchain-Based Electronic Laboratory Notebook. Med - Clin - Res. 2026;2(8):105-109. DOI: <https://www.doi.org/10.52106/3069-9088.1030>.

Copyright

© 2026 Authors. This is an open- access article distributed under the terms of the Creative Commons Attribution 4.0 International license.

No.	PubMed ID	Requirement(s) for ELN
1	19487210 [15]	(a) Capture information as experiments are executed. (b) Annotations between or across experiments. (c) Consistent scientific terminology.
2	21702721 [16]	(a) Interface with other computer systems and instruments. (b) Highly configurable. (c) Validated templates for data input. (d) Input via barcode scanning or dropdown menus for clarity and consistency. (e) Capture diverse file types. (f) Real-time error checking. (g) Secured data access.
3	23360752 [17]	(a) Security and logging. (b) Automatic archival. (c) Capture diverse file types.
4	24423594 [18]	(a) Comply with 21 CFR part 11. (b) Comply with required GLP regulations. (c) Interface with dossier preparation systems. (d) Ensure data integrity. (e) Checking and auditing electronic data
5	26835004 [19]	(a) Comply with 21 CFR part 11. (b) User rights management. (c) Interface with other computer systems and instruments.
6	29086051 [20]	(a) Search existing records. (b) Data sharing. (c) Capture diverse file types. (d) Secured data access. (e) Multi-location data access. (f) Secured data backup. (g) Improved record keeping. (h) Improved group and project management.
7	29308130 [21]	(a) Worldwide data sharing. (b) User management. (c) Multi-location data access. (d) Revision / version tracking. (e) Providence of data. (f) Collaboration and discussion among users. (g) Search existing records. (h) Project record with diverse files.
8	29653352 [22]	(a) Archival. (b) Calendar. (c) User management. (d) Notifications. (e) Search.
9	30900066 [23]	(a) Tagging records. (b) Search existing records.
10	41857028 [24]	Data should be stored in a FAIR way: (a) findable, (b) accessible, (c) interoperable, and (d) reusable.
11	42124799 [25]	Data and processes should be stored in a FAIR way: (a) findable, (b) accessible, (c) interoperable, and (d) reusable.

usage; as follow:

1. System and Compliance

Highly configurable.
User / group / project rights management.
Checking and auditing electronic data
Comply with 21 CFR part 11, and GLP regulations.

2. Data capture

Capture data as experiments is executed for providence.
Validated templates for data input.
Input via barcode scanning or dropdown menus for clarity and consistency.
Capture diverse file types.
Real-time error checking.
Revision / version tracking to ensure data integrity.

3. Data usage

Secured data access.
Secured multi-location data access.
Tag and search existing records (findable and accessible) with consistent scientific terminology.
Secured data backup.
Collaboration and discussion among users.
Calendar and notifications.
Interface with other computer systems (interoperable and reuseable), such as dossier preparation systems; and instruments.

Blockchain and Its Potential Implications in ELN

Blockchain is a distributed digital ledger technology that records transactions as an ordered chain of cryptographically linked blocks [26]. Each block contains a set of records together with a cryptographic hash of the previous block, making it computationally difficult to alter past entries without detection [27,28]. In the context of ELN, blockchain can complement by providing tamper-resistant, and auditable digital record [29]. For example, cryptographic hashes of notebook entries, datasets, protocols, or analysis results can be recorded on a blockchain without exposing their actual contents; thereby, preserving confidentiality while establishing

proof of existence and detecting unauthorized modifications. This capability can strengthen research integrity, support regulatory compliance, facilitate intellectual property claims, and enhance trust in collaborative or multi-institutional research. Moreover, the search term, blockchain AND ("electronic laboratory notebook" OR "electronic laboratory notebooks" OR "electronic lab notebook" OR "electronic lab notebooks"), had only one hit on 30 June 2026 using PubMed (URL #2) – IntegriLAB [29]. This suggests that the use of blockchain technology on ELN is in its infancy.

Requirements for a Blockchain-Based ELN

The proposed blockchain-based ELN should retain all conventional ELN requirements identified in the scoping review. Blockchain should be implemented as an additional integrity layer rather than as a replacement in areas but not limited to areas like data-entry interfaces, search, instrument integration, file repositories, or conventional database services.

A blockchain-based ELN should consist of five principal components:

1. An ELN application layer through which researchers enter, review, search, and manage experimental records.
2. An encrypted off-chain repository for raw data, images, documents, analytical code, and other large or confidential files crucial in ensuring data integrity.
3. A blockchain integrity layer containing cryptographic hashes and the minimum metadata required to verify provenance.
4. Key identity or management record service for authentication, namely digital signatures, roles, relation to assignment, key rotation, and revocation.
5. An interoperability layer connecting laboratory instruments, laboratory data repositories, analytical software, systems, and publications through documented application programming interfaces.

This hybrid architecture avoids placing complete research datasets directly on the blockchain. It also enables large files to be stored efficiently while preserving the ability to verify whether an off-chain

file is identical to the version originally registered. A 28-domain guideline is proposed to maintain the integrity of the ELN:

1. **Conventional ELN functions:** The system should support the core conventional ELN functions identified in Table 1 with regulatory and compliance applied to the system intended use.
2. **Capture at source:** Should raw data be possible to be recorded directly from instruments or analytical systems, reducing dependence on manual transcription.
3. **Contextual metadata:** Each record must include sufficient metadata to identify key details like the project, experiment, operator, instrument, sample, protocol, software version, date, and time.
4. **Cryptographic hashing:** A cryptographic hash must be generated when a record or file is first captured. When an approved hashing algorithm is verified by auditors or superiors the version must be documented.
5. **Integrity verification:** The system must allow an authorized user or external auditor to verify by recalculation and comparing the data file registered on the blockchain.
6. **Digital signatures:** Creation, review, approval, and correction of records must be linked to authenticated digital signatures that cannot be transferred to another record.
7. **Trusted timestamps:** Transactions must use synchronized and auditable time zone sources. The system must document the applicable time zone and prevent ordinary users from modifying timestamps.
8. **Append-only versioning:** Zero tolerance of silent overwrite of any existing records. A correction must create a new version that refers to the previous version and records the reason, author, date, and time of the change.
9. **Provenance and chain of custody:** Traceability must not be compromised and diligently documented in the lifecycle of research events; primarily data creation, transfer, transformation, analysis, review, approval, export, and publication.
10. **Hybrid storage:** Large files and substantive research data must normally be stored off-chain. The blockchain should store their hashes, timestamps, version identifiers, and minimal provenance metadata.
11. **Confidentiality:** Sensitive scientific, personal, commercial, and intellectual-property information must not be written to a publicly accessible ledger. Data must be encrypted in transit and at rest.
12. **Privacy by design:** The system must minimize personal information recorded on the blockchain and must define procedures for rectification, withdrawal, retention, and deletion of off-chain information.
13. **Permissioned participation:** A permissioned blockchain should be used where records involve confidential or regulated research. Only approved organizations and authenticated users should be permitted to submit or validate transactions.
14. **External independent anchor / verification:** Where stronger external verification is required, the system may periodically register a summary hash or Merkle root from the permissioned ledger on an independently operated public blockchain.
15. **Role-based access:** Access must be assigned according to clear distinctive roles in the project such as researcher, principal investigator, and external collaborator.
16. **Separation of duties:** The same user should not be able to create, approve, and administratively alter the configuration governing a record.
17. **Key security management:** The system must provide secure key generation, multifactor authentication, storage security and recovery, and procedures for compromised or lost credentials.
18. **Smart-contract controls:** Smart contracts may enforce

workflow requirements, but their source code, versions, testing records, and approval history must be retained and auditable.

19. **Regulatory compliance:** When applicable, the system must be validated for its intended use, support relevant requirements and regulation like under 21 CFR Part 11, GLP, and institutional policies.
20. **Audit presentation:** The system must generate complete auditable accurate, human and machine-readable copies of records, signatures, and audit histories.
21. **FAIR and interoperable data:** Records where possible should use standardized metadata, controlled scientific terminology, persistent identifiers, documented APIs, and non-proprietary export formats.
22. **Availability and recovery:** Blockchain nodes, off-chain data, encryption keys, and system configuration must be covered by backup, disaster-recovery, business-continuity, and restoration-testing procedures.
23. **Retention and migration:** The system must preserve the ability to verify records throughout the required retention period, including after software upgrades, cryptographic changes, or migration to another platform.
24. **Scalability:** The architecture must be tested using realistic file sizes, transaction frequencies, user numbers, and instrument workloads. Transactions may be batched where individual blockchain registration is unnecessary.
25. **Security monitoring:** The system must support vulnerability management, security logging, incident detection, penetration testing, and investigation of unauthorized transactions or compromised nodes.
26. **Network governance:** Written governance must define node ownership, validator responsibilities, consensus rules, onboarding and removal of participants, software upgrades, disputes, and system termination.
27. **Usability:** Blockchain operations should occur automatically in the background and should not substantially increase the documentation burden on laboratory personnel.
28. **Transparency of assurance:** The interface must clearly state that blockchain verification demonstrates record integrity and provenance after registration, but does not prove the scientific accuracy of the original observation.

Concluding Remarks

Existing ELNs provide important functions for experimental documentation, file management, collaboration, search, and regulatory recordkeeping. Nevertheless, their audit histories frequently remain dependent on a centrally controlled system. Blockchain technology may supplement an ELN by providing cryptographically verifiable timestamps, digital signatures, append-only version histories, and distributed validation of selected research events. A practical blockchain-based ELN should use a hybrid architecture in which substantive data are stored securely off-chain while hashes and minimal provenance metadata are registered on a permissioned blockchain. The system must also provide strong identity management, access control, privacy protection, interoperability, regulatory validation, long-term archival, disaster recovery, and clearly defined network governance. However, blockchain should not be presented as a complete solution to data fabrication, falsification, or poor scientific practice. It cannot determine whether the information registered at its source is accurate. Its more defensible role is to provide evidence of provenance and to make subsequent alteration detectable. When implemented selectively within a broader framework of scientific, procedural, and institutional controls, blockchain may strengthen the traceability, reproducibility, and trustworthiness of electronic laboratory records

URL

URL #1: [https://pubmed.ncbi.nlm.nih.gov/?term=\(require*+OR+consider*\)+AND+\(\"electronic+laboratory+notebook\"+OR+\"electronic+laboratory+notebooks\"+OR+\"electronic+lab+notebook\"+OR+\"electronic+lab+notebooks\"\)&filter=dates.1000/1/1-2026/06/30](https://pubmed.ncbi.nlm.nih.gov/?term=(require*+OR+consider*)+AND+(\)

URL #2: [https://pubmed.ncbi.nlm.nih.gov/?term=blockchain+AND+\(\"electronic+laboratory+notebook\"+OR+\"electronic+laboratory+notebooks\"+OR+\"electronic+lab+notebook\"+OR+\"electronic+lab+notebooks\"\)&filter=dates.1000/1/1-2026/06/30](https://pubmed.ncbi.nlm.nih.gov/?term=blockchain+AND+(\)

Supplementary Materials

Supplementary materials can be downloaded from <https://bit.ly/BlockchainELN>.

References

- Fang FC, Casadevall A. Retracted Science and The Retraction Index. *Infect Immun*. 2011; 79(10): 3855-3859. DOI: 10.1128/IAI.05661-11.
- Fang FC, Steen RG, Casadevall A. Misconduct Accounts for the Majority of Retracted Scientific Publications. *Proc Natl Acad Sci U S A*. 2012; 109(42): 17028-17033. DOI: 10.1073/pnas.1212247109.
- Lievore C, Rubbo P, Dos Santos CB, Picinin CT, Pilatti LA. Research Ethics: A Profile of Retractions from World Class Universities. *Scientometrics*. 2021; 126(8): 6871-6889. DOI: 10.1007/s11192-021-03987-y.
- Faggion CM, Ware RS, Bakas N, Wasiak J. An Analysis of Retractions of Dental Publications. *J Dent*. 2018; 79: 19-23. DOI: 10.1016/j.jdent.2018.09.002.
- Fanelli D. How Many Scientists Fabricate and Falsify Research? A Systematic Review and Meta-Analysis of Survey Data. *PLOS ONE*. 2009; 4(5): e5738. DOI: 10.1371/journal.pone.0005738.
- Ling MH. Science/Education Portraits III: Perceived Prevalence of Data Fabrication and/or Falsification in Research. *Advances in Biotechnology and Microbiology*. 2018; 11(5): 555824.
- Phogat R, Manjunath BC, Sabbarwal B, Bhatnagar A, Reena null, et al. Misconduct in Biomedical Research: A Meta-Analysis and Systematic Review. *J Int Soc Prev Community Dent*. 2023; 13(3): 185-193. DOI: 10.4103/jispcd.JISPCD_220_22.
- Van Steensel B. Scientific Honesty and Publicly Shared Lab Notebooks: Sharing Lab Notebooks Along with Publication Would Increase Transparency and Help to Improve Honesty When Reporting Results. *EMBO Rep*. 2018; 19(10): e46866. DOI: 10.15252/embr.201846866.
- Higgins SG, Nogiwa-Valdez AA, Stevens MM. Considerations for Implementing Electronic Laboratory Notebooks in an Academic Research Environment. *Nat Protoc*. 2022; 17(2): 179-189. DOI: 10.1038/s41596-021-00645-8.
- Machina HK, Wild DJ. Electronic Laboratory Notebooks: Progress and Challenges in Implementation. *Journal of Laboratory Automation*. 2013; 18(4): 264-268. DOI: 10.1177/2211068213484471.
- Ling MH. NotaLogger: Notarization Code Generator and Logging Service. *The Python Papers*. 2013; 9(1): 2.
- Fleming EJ. An Approach to Improve Laboratory Notebook Feedback and Decrease Instructor Fatigue from Grading Undergraduate Laboratory Notebooks. *Journal of Microbiology & Biology Education*. 2023; 24(1): e00018-22. DOI: 10.1128/jmbe.00018-22.
- Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, et al. The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ*. 2021; 372: n71. DOI: 10.1136/bmj.n71.
- Sarkis-Onofre R, Catalá-López F, Aromataris E, Lockwood C. How to properly use the PRISMA Statement. *Systematic Reviews*. 2021; 10(1): 117, s13643-021-01671-z. DOI: 10.1186/s13643-021-01671-z.
- Martin CJ, Haji MH, Jimack PK, Pilling MJ, Dew PM. A User-Orientated Approach to Provenance Capture and Representation for In Silico Experiments, Explored Within the Atmospheric Chemistry Community. *Philosophical Transactions Series A, Mathematical, Physical, and Engineering Sciences*. 2009; 367(1898): 2753-2770. DOI: 10.1098/rsta.2009.0044.
- Beato B, Pisek A, White J, Grever T, Engel B, et al. Going Paperless: Implementing an Electronic Laboratory Notebook in a Bioanalytical Laboratory. *Bioanalysis*. 2011; 3(13): 1457-1470. DOI: 10.4155/bio.11.117.
- Rees I, Langley E, Chiu W, Ludtke SJ. EMEN2: An Object Oriented Database and Electronic Lab Notebook. *Microscopy and Microanalysis: The Official Journal of Microscopy Society of America, Microbeam Analysis Society, Microscopical Society of Canada*. 2013; 19(1): 1-10. DOI: 10.1017/S1431927612014043.
- McDowall RD. Developing a Strategy for a Regulated Electronic Bioanalytical Laboratory. *Bioanalysis*. 2014; 6(2): 165-184. DOI: 10.4155/bio.13.282.
- Dirnagl U, Przesdzin I. A Pocket Guide to Electronic Laboratory Notebooks in the Academic Life Sciences. *F1000Res*. 2016; 5: 2. DOI: 10.12688/f1000research.7628.1.
- Kanza S, Willoughby C, Gibbins N, Whitby R, Frey JG, et al. Electronic Lab Notebooks: Can They Replace Paper?. *J Cheminform*. 2017; 9(1): 31. DOI: 10.1186/s13321-017-0221-3.
- Badiola KA, Bird C, Brocklesby WS, Casson J, Chapman RT, et al. Experiences with a Researcher-Centric ELN. *Chemical Science*. 2015; 6(3): 1614-1629. DOI: 10.1039/c4sc02128b.
- Rahman M, Watabe H. Online Molecular Image Repository and Analysis System: A Multicenter Collaborative Open-Source Infrastructure for Molecular Imaging Research and Application. *Computers in Biology and Medicine*. 2018; 96: 233-240. DOI: 10.1016/j.combiomed.2018.04.001.
- Kanza S, Gibbins N, Frey JG. Too Many Tags Spoil the Metadata: Investigating the Knowledge Management of Scientific Research with Semantic Web Technologies. *Journal of Cheminformatics*. 2019; 11(1): 23. DOI: 10.1186/s13321-019-0345-8.
- Kirchner F, Wieland DCF, Irvine S, Schimek S, Reimers J, et al. An Ontology-Based Description of Nano Computed Tomography Measurements in Electronic Laboratory Notebooks. *Scientific Data*. 2026; 13(1): 432. DOI: 10.1038/s41597-026-07052-2.
- Yakutovich AV, Hollas D, Bainglass E, Yu J, Battaglia C, et al. Accelerating discovery across scientific disciplines through reproducible workflows with AiDALab. *Digit Discov*. 2026; 5(5): 2310-2324. DOI: 10.1039/d5dd00567a.
- Habib G, Sharma S, Ibrahim S, Ahmad I, Qureshi S, et al. Blockchain Technology: Benefits, Challenges, Applications, and Integration of Blockchain Technology with Cloud Computing. *Future Internet*. 2022; 14(11): 341. DOI: 10.3390/fi14110341.
- Ling MH. SEcured REcorder BOx (SEREBO) based on blockchain technology for immutable data management and notarization. *MOJ Proteomics & Bioinformatics*. 2018; 7(6): 169-174.

28. Dong S, Abbas K, Li M, Kamruzzaman J. Blockchain technology and application: an overview. *PeerJ Computer Science*. 2023; 9: e1705. DOI: 10.7717/peerj-cs.1705.
29. Easmin R, Veronese M, Allen P, Nordio G, Turkheimer F. IntegriLAB: a blockchain-enabled electronic lab notebook for reproducible neuroimaging research. *Front Neuroinform*. 2026; 20: 1753937. DOI: 10.3389/fninf.2026.1753937.